AQ10.4: Activity Questions 4 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Let $f(x, y)$f(x, y) be defined in a neighbourhood of (a, b)(a, b). When is (a, b)(a, b) a critical point of ff?

- [ ] $f(a, b) = 0$f(a, b) = 0.
- [ ] $f_x(a, b) = 0$fx(a, b) = 0.
- [ ] $f_x(a, b) = 0 = f_y(a, b)$fx(a, b) = 0 = fy(a, b).
- [ ] f_x fx does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x(a, b) = 0 = f_y(a, b)$fx(a, b) = 0 = fy(a, b).

f_x fx does not exist.

1 point

Consider $f(x, y) = \sqrt{5x^2 + 3y + 2}$f(x, y) = $\sqrt{\quad 5x2 + 3y + 2 \quad}$.

- [ ] Any point on the line $x + y = 1$x + y = 1 in the domain of definition of ff satisfies $f_y = 0$fy = 0.
- [ ] There is no (x, y)(x, y) such that $f_y = 0$fy = 0.
- [ ] Solutions of $5x^2 + 3y + 2 = 0$5x2 + 3y + 2 = 0 are critical points of ff.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There is no (x, y)(x, y) such that $f_y = 0$fy = 0.

Solutions of $5x^2 + 3y + 2 = 0$5x2 + 3y + 2 = 0 are critical points of ff.

1 point

A critical point

- [ ] is a point of maximum.
- [ ] may be a saddle point.
- [ ] may be a point of local minimum.
- [ ] can be obtained by equating f_x fx and f_y fy to 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

may be a saddle point.

may be a point of local minimum.

can be obtained by equating f_x fx and f_y fy to 0.

1 point

Let $f(x, y) = x^2\, y$f(x, y) = x2y. Then

- [ ] (0, 0)(0, 0) is a critical point.
- [ ] (0, y)(0, y) for $y \in R$y ∈ R are critical points.
- [ ] f_x > 0fx > 0 at the critical points.
- [ ] f_y = 0fy = 0 at the critical points.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(0, 0)(0, 0) is a critical point.

(0, y)(0, y) for $y \in R$y ∈ R are critical points.

f_y = 0fy = 0 at the critical points.

1 point

For $f(x, y) = x^3 + 2x\, y - 2x - 4y$f(x, y) = x3 + 2xy − 2x − 4y, which of the following is/are true?

- [ ] (2, − 5)(2, −5) is a critical point.
- [ ] (2, 5)(2, 5) is a critical point.
- [ ] f_y = 0fy = 0 is a straight line parallel to the yy-axis.
- [ ] The system of equations f_x = 0fx = 0, f_y = 0fy = 0 has a unique solution.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(2, − 5)(2, −5) is a critical point.

f_y = 0fy = 0 is a straight line parallel to the yy-axis.

The system of equations f_x = 0fx = 0, f_y = 0fy = 0 has a unique solution.

1 point

Choose the correct statement(s) about the function $f(x, y) = x\, y^2 + 2x$f(x, y) = xy2 + 2x.

- [ ] f_x = 0fx = 0 only for 2 values of xx.
- [ ] f_y = 0fy = 0 along both xx and yy axes.
- [ ] There are no critical points of ff.
- [ ] The point (0, 0)(0, 0) is a point of local minimum.

No, the answer is incorrect.

Score: 0

Accepted Answers:

f_y = 0fy = 0 along both xx and yy axes.

There are no critical points of ff.

1 point

For $f(x, y) = \frac{x^3}{3} + y^2 + 2xy - 6x - 3y + 4$f(x, y) = 3x3 + y2 + 2xy − 6x − 3y + 4

- [ ] $f_y = 0$fy = 0 is a straight line passing through the origin.
- [ ] $(-1, \frac{5}{2})$(−1, 25) and $(3, -\frac{3}{2})$(3, −23) are critical points.
- [ ] $f_x(-1, \frac{5}{2}) = 0$fx(−1, 25) = 0.
- [ ] $f_y(3, -\frac{3}{2}) = 2$fy(3, −23) = 2.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(-1, \frac{5}{2})$(−1, 25) and $(3, -\frac{3}{2})$(3, −23) are critical points.
$f_x(-1, \frac{5}{2}) = 0$fx(−1, 25) = 0.

1 point
Choose the correct statement(s) about $f(x, y) = x^4 + y^4$ f(x, y) = x4 + y4.

- [ ] There are no critical points.
- [ ] There is exactly one critical point.
- [ ] (0, 0)(0, 0) is a point of global minimum.
- [ ] (0, 0)(0, 0) is a point of local minimum.

No, the answer is incorrect.
Score: 0
Accepted Answers:
There is exactly one critical point.
(0, 0)(0, 0) is a point of global minimum.
(0, 0)(0, 0) is a point of local minimum.

Level 2

1 point
Let $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$f(x, y) = x2 − 2xy + 4y2 − 4x − 2y + 2. Then

- [ ] (3, 1)(3, 1) is a critical point.
- [ ] (2, 1)(2, 1) is a critical point.
- [ ] Value of ff at (2, 1)(2, 1) is -4.
- [ ] Value of ff at (3, 1)(3, 1) is -9.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(3, 1)(3, 1) is a critical point.
Value of ff at (2, 1)(2, 1) is -4.

1 point
Let $f(x, y) = x^2 - 2xy + 4y^2 - 4x - 2y + 2$f(x, y) = x2 − 2xy + 4y2 − 4x − 2y + 2 be defined on the box $0 \le x \le 2$0 ≤ x ≤ 2 and $0 \le y \le 1$0 ≤ y ≤ 1. Then which of the following options is (are) true?

- [ ]

Along the line joining $(0, 0)(0, 0)$ and $(2, 0)(2, 0)$, the minimum is attained at $(2, 0)(2, 0)$.

$(0, 1)(0, 1)$ is a point of extremum along the line joining $(0, 1)(0, 1)$ and $(2, 1)(2, 1)$.

$f(0, 1) = 1$f(0, 1) = 1

None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Along the line joining $(0, 0)(0, 0)$ and $(2, 0)(2, 0)$, the minimum is attained at $(2, 0)(2, 0)$.

$(0, 1)(0, 1)$ is a point of extremum along the line joining $(0, 1)(0, 1)$ and $(2, 1)(2, 1)$.

1 point

Let $f(x, y) = sin\ x\ cos\ y$f(x, y) = sin x cos y. Then

$((2k + 1)\frac{\pi}{2}, k\pi)((2k + 1)2\pi, k\pi)$ for $k \in N$k ∈ N are critical points of ff.

$((2k + 1)\frac{\pi}{2}, (2k + 1)\frac{\pi}{2})((2k + 1)2\pi, (2k + 1)2\pi)$ for $k \in N$k ∈ N are critical points of ff.

$(k\pi, k\pi)(k\pi, k\pi)$ for $k \in N$k ∈ N are critical points of ff.

$(k\pi, (2k + 1)\frac{\pi}{2})(k\pi, (2k + 1)2\pi)$ for $k \in N$k ∈ N are critical points of ff.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$((2k + 1)\frac{\pi}{2}, k\pi)((2k + 1)2\pi, k\pi)$ for $k \in N$k ∈ N are critical points of ff.

$(k\pi, (2k + 1)\frac{\pi}{2})(k\pi, (2k + 1)2\pi)$ for $k \in N$k ∈ N are critical points of ff.

1 point

Let $f(x, y) = e^{-x^2 + y - \frac{y^3}{3}}$f(x, y) = e−x2+y−3y3. Then

$f_x(1, 0) = -\frac{2}{e}$fx(1, 0) = −e2.

$f_x(0, 0) = 1$fx(0, 0) = 1.

$f_y(0, -1) = 0$fy(0, −1) = 0.

$f_y(1, 0) = e$fy(1, 0) = e.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f_x(1, 0) = -\frac{2}{e}$fx(1, 0) = −e2.

$f_y(0, -1) = 0$fy(0, −1) = 0.

1 point

Let $f(x, y) = e^{-x^2 + y - \frac{y^3}{3}}$f(x, y) = e−x2+y−3y3. Then

$(0, 1)(0, 1)$ is a critical point.

$(0, -1)(0, −1)$ is a critical point.

$(1, 1)(1, 1)$ is a critical point.

☐

$(1, -1)(1, -1)$ is a critical point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(0, 1)(0, 1)$ is a critical point.

$(0, -1)(0, -1)$ is a critical point.

1 point

Find three positive numbers whose sum is 12 such that the sum of their squares is minimum. The mathematical expression that best defines this problem is

○ minimize $x^2 + y^2 + z^2$ x2 + y2 + z2 such that $x + y + z = 12$, $x, y, z > 0$x, y, z > 0.

○ minimize $x + y + z$ x + y + z such that $x^2 + y^2 + z^2 = 12$x2 + y2 + z2 = 12, $x, y, z \geq 0$x, y, z ≥ 0.

○ minimize $x^2 + y^2 + z^2$ x2 + y2 + z2 such that $x + y + z = 12$x + y + z = 12, $x, y, z \geq 0$x, y, z ≥ 0.

○ minimize $x + y + z$ x + y + z such that $x^2 + y^2 + z^2 = 12$x2 + y2 + z2 = 12, $x, y, z \geq 0$x, y, z ≥ 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

minimize $x^2 + y^2 + z^2$ x2 + y2 + z2 such that $x + y + z = 12$x + y + z = 12, $x, y, z > 0$x, y, z > 0.

Check Answers

Your score is: 0/14